SpeechLens

Contrastive speech analytics with temporal flaw grounding · Multimodal AI Hackathon 2026, Track C · Technical documentation

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SpeechLens compares a participant's delivery of a text with a well-delivered reference reading of the *same* text. It force-aligns both readings word by word, measures how pace, pausing, pitch, loudness, articulation and fluency differ, marks each delivery flaw with start and end times, and explains it with the measured numbers and a fix. A rubric turns the findings into a reproducible 0–100 score. It runs on a CPU with no language model and gives byte-identical output on every run. We built the contrastive dataset it is trained and tested on, and every reported number is cross-validated.

0.66
F1 locating 11 flaw types (95 % CI 0.62–0.70), precision 0.82, leave-one-speech-out

150 ms
median boundary error of located flaws; mean tIoU 0.76

0.03 /min
false findings on unflawed controls, incl. voice ±4 semitones

ρ = −0.95
score vs. flaw level across the mixed-flaw gradient

91 %
of findings vanish when only the named cause is repaired (counterfactual test)

1. Dataset construction

Gold baseline

Five celebrated public-domain speeches (US federal government works), audio and transcripts from the Miller Center, University of Virginia. A greedy CTC decode locates the speech inside each recording, skipping oaths and introductions, and an excerpt of about 75 s is cut at a sentence boundary.

Speech	Len (s)	Words	Align conf.
Roosevelt, Day of Infamy (1941)	110.5	158	0.91
Kennedy, Inaugural (1961)	79.1	131	0.94
Johnson, We Shall Overcome (1965)	88.9	120	0.95
Reagan, Challenger (1986)	77.2	197	0.92
Obama, Inaugural (2009)	81.4	126	0.94

Align conf. = mean CTC posterior of the transcript's characters over each word.

The bad spectrum: Flaw Injection Engine

Each gold recording is cut at the midpoints of inter-word gaps and one span is edited with Praat PSOLA resynthesis, which changes timing and pitch while keeping the speaker's own voice. Because we make every edit, each label is exact to the sample: new word times are computed by integrating the same piecewise-linear duration tier Praat applies, and inserted material shifts later times exactly. Eleven flaw types at five severities, from near-perfect (1) to egregious (5); the last two are reading errors, where the participant departs from the text itself:

Flaw	Edit	Severity 1 → 5
Rushed / dragging	duration tier; words ×f, pauses $\times f^{1.6}$	f 0.88 → 0.55 / 1.12 → 1.62
Monotone	pitch excursion around span median ×(1−a)	a 35 % → 97 %
Trailing off	gain ramp over 60 % of span	−4 → −18 dB
Missing pauses	punctuation pauses cut, centre kept	60 % → 4 % kept
Awkward pause	mid-phrase silence with the recording's own noise floor	0.35 → 1.8 s
Filler (uh/um)	speaker's steadiest vowel lengthened by overlap-add, falling pitch, nasal tail for "um"	0.28 → 0.65 s, 1–4×
Muffled articulation	6th-order low-pass	5 → 1.2 kHz
Repeated onset	word's first 150 ms repeated	1 → 3×
Skipped words	words cut from mid-gap to mid-gap	1 → 3 words, 1–2×
Restarted phrase	words said again after a short pause	1 → 4 words, 1–3×

Per speech: 55 single-flaw clips, 8 mixed clips forming a gradient from two mild to six severe flaws, 3 clips combining a reading error with two prosodic flaws, and 4 **controls** with no flaw (untouched, voice shifted +4 and −4 semitones, 12 dB quieter). All random choices are seeded from a CRC32 of the clip id, and Praat's own random generator is seeded per call, so a rebuild is bit-identical on any machine. Total: 350 synthetic clips. A datasheet (`docs/DATASHEET.md`) documents composition, collection and known gaps.

Human recordings and manipulation check

Three team members each read three excerpts four times: a clean take A and three takes with flaws performed at positions fixed by a recording script (B pace, C voice, D fluency): 36 recordings, 90 scripted flaws. Labels come from force-aligning each take and mapping the scripted words to times. Each flawed take is compared with the *same person's* clean take, since a student's reading against President Kennedy's differs everywhere.

A **manipulation check** accepts a scripted flaw only if a fixed raw measurement over the scripted words, against take A, reaches the severity-1 injection level (e.g. duration factor ≤ 0.88 for rushed, pitch range ratio ≤ 0.65 for monotone). The measurements do not use the detector. **58 of 90** passed. Phone-recorder gain control erased most fades (2/9), and "mumbles" were usually whispered, which raises high-frequency energy instead of lowering it (2/9). Unperformed labels are excluded from scoring; findings that match them are neither rewarded nor penalised.

Files: `data/library/<id>/{audio.wav, transcript.txt, alignment.json}`, `data/dataset/{clips,labels}/`, `manifest.json`. Each label: type, severity, start, end (s), word span, edit parameters.

2. Architecture

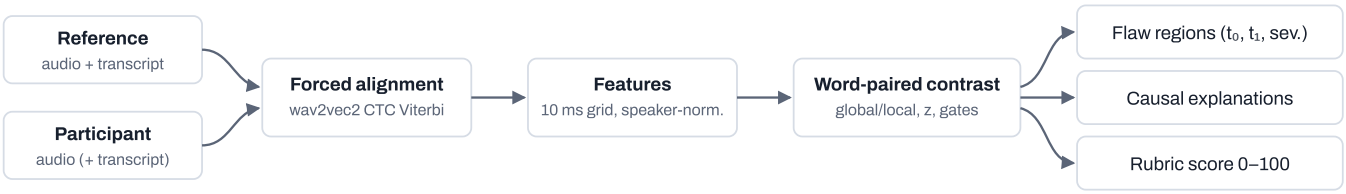


Figure 1. Pipeline. Both readings go through the same alignment and feature extraction; findings come only from their difference.

2.1 Forced alignment

`wav2vec2-base-960h` (ONNX, CPU) outputs per-frame log-probabilities *L[t,c]* over 32 symbols at a 20 ms stride; long audio is processed in 20 s windows with 1 s context so results do not depend on length. Transcripts are normalized (numbers and years to words, "December 7" to "seventh", dashes split). An exact CTC Viterbi pass finds the single path spelling the transcript:

$$\alpha_t(s) = L[t, \text{ext}_s] + \max(\alpha_{t-1}(s), \alpha_{t-1}(s-1), \alpha_{t-1}(s-2) \text{ if } \text{ext}_s \neq \text{blank}, \text{ext}_s \neq \text{ext}_{s-2})$$

Word boundaries are then snapped to the 10 ms RMS envelope (−32 dB re. 95th percentile), extending CTC's spiky onsets to cover each word's decaying tail, so pauses are measured from real silence.

2.2 Feature extraction

- **F0**: Praat autocorrelation, two passes with a speaker-adaptive range (floor 0.75×Q25, ceiling 1.5×Q75 of pass one).
- **Spectrum**: 512-point FFT (STFT, 10 ms hop): share of energy above 2 kHz and 4 kHz (consonant articulation), spectral flux, centroid, 13 MFCCs from 40 mel bands.
- **Intensity, HNR** (Praat); per word: articulation rate (syllables / word duration, pauses excluded), pause after, voiced sound inside the pause, pitch p10–p90 range, loudness, recognizer confidence.

Voice quality (descriptive, not scored): jitter and shimmer (local, Praat point process), HNR and smoothed cepstral peak prominence for both readings, shown side by side.

Speaker-agnostic normalization. Pitch is in semitones relative to each speaker's own median, and loudness in dB relative to the speaker's median speech level:

$$\text{st}(t) = 12 \cdot \log_2(\text{F0}(t) / \text{median F0}) \qquad \text{I}_{\text{rel}}(t) = \text{I}(t) - \text{median}_{\text{speech}}\ \text{I}$$

A deep and a high voice reading identically give the same contours; recording gain cancels. The controls test this directly: shifting the voice ±4 semitones or lowering it 12 dB produced 1 finding in 29 minutes.

3. Temporal grounding and causal explanation

Pairing. Words of the two readings are matched by transcript position (sequence alignment of normalized words), so a skipped word affects only itself.

Global vs local. For each contrast signal s_j over word pairs (e.g. $\log_2$ of the rate ratio over a 3-word window), the median is the speaker's overall habit and is reported separately; the residual is where delivery departs from the reference, scored against the reference speaker's own word-to-word variability:

$$g = \text{median}_j\ s_j, \quad l_j = \text{smooth}_3(s_j - g), \quad z_j = l_j / (1.4826 \cdot \text{MAD}(\text{reference}))$$

Runs of words past a threshold (one-word holes closed) become regions: rushed/dragging ($|\log_2$ rate ratio$| >$ 0.20, ≥3 words, change spread across the words), monotone (pitch-range ratio < 0.68 over 5-word

windows, octave errors removed), trailing off (> 3.5 dB below reference), muffled (mean $\log_2$ ratio of >2 kHz and >4 kHz energy < −0.75).

Disfluencies at word boundaries. A disfluency adds time at one spot; a slow passage adds it everywhere. Each boundary's added time is measured beyond the 75th percentile of the six neighbouring boundaries. A *repeated onset* is ≥2 short fragments before a word whose first fragment matches the word's own onset (MFCC cosine > 0.88 for one repetition). A *filler* is a held vowel (steady pitch ±0.5 st, calmest 30 % of spectral change) ≥140 ms longer than the reference holds there. An *awkward pause* is ≥0.3 s extra mid-phrase; at punctuation only ≥1.2 s counts, since a longer pause there is usually deliberate. *Missing pauses* means ≥2 reference pauses within 8 words cut below 40 %.

Reading errors. Forced alignment must place every transcript word somewhere: a word never said is squeezed into a few frames with a near-zero posterior (< 0.10, the reference ≥ 0.50, duration < 55 % of the reference's) and becomes a *skipped-word* candidate. A restart ("we choose to — we choose to go") is extra time at one boundary that holds speech. For both, a greedy CTC recognition pass over the 2–5 s neighbourhood with the same acoustic model confirms the event: the skipped words must be absent, and a restart is a word sequence recognised twice in a row that the transcript does not repeat (so "we shall fight, we shall fight" in the text itself is never flagged). The recognised text becomes the finding's evidence ("recognised here: to our secretary to our secretary of state").

Conflict resolution. A short span deviation around a point disfluency is folded into it (the aligner absorbs an "uh" into the next word, which then looks slow); missing pauses inside a rushed span, and stretched pauses inside a dragging span, belong to that span.

Significance gates. Natural rereading drifts by about ±20 % in pace and hundreds of milliseconds in pauses. A per-type magnitude gate, fitted to maximise F1 on synthetic and human data (each weighted half), drops findings within that variation, e.g. rushed only below a duration factor of 0.73, filler only above 0.38 s.

Causal explanation. Each region fills a fixed template with its measurements: location and words, primary metric for reference vs participant with the relative change, z-score, the acoustic or perceptual rationale, and a fix. No language model is involved, so the same audio always gives the same sentence.

> From 10.61 s to 11.88 s (“Today is a day for mourning and”), articulation rate was 7.09 syll/s against 5.06 syll/s in the reference (+40 %); that is 1.6× the reference speaker's natural word-to-word variation. Faster articulation compresses the stressed syllables and the micro-pauses listeners use to segment phrases… Fix: slow to the reference pace here and let the pauses between clauses breathe.

4. Rubric and scoring

Six dimensions (pace, pausing, fluency, pitch variety, volume control, clarity) start at 10. A span flaw costs severity × (0.8 + 6 × share of words covered), a point flaw 0.7 × severity, a whole-reading habit 2.4 × severity. The overall score is 10 × the weighted mean of the dimensions; presets weight them for Declamation (pitch variety, pausing), Interpretive Reading (pitch variety ×1.5), Extemporaneous (fluency ×1.5) and Persuasive Oratory (volume ×1.4), with bands Championship ≥ 90, Strong ≥ 75, Developing ≥ 60.

Severity is read from each region's raw magnitude through a table fitted from data (median magnitude per true severity, monotone). On speeches held out of the fit, the estimate is off by 0.49 levels on average and within one level 93 % of the time (0.60 levels with the injection parameters as the table).

5. Evaluation

Every number runs the full pipeline on raw audio, alignment of the flawed recording included. A finding matches a label of the same type at temporal IoU ≥ 0.3, one to one. Thresholds and severity tables are fitted from data, so results are cross-validated: synthetic clips are scored with gates fitted without that speech (and without human takes of it); human takes with gates fitted without that recorder.

Flaw	P	R	F1	Human R
Rushed	0.78	0.70	0.74	4/8
Dragging	0.82	0.52	0.64	5/5
Monotone	1.00	0.61	0.76	2/4
Trailing off	0.93	0.63	0.75	2/2
Missing pauses	0.96	0.50	0.66	4/8
Awkward pause	0.87	0.59	0.70	5/7
Filler	0.72	0.62	0.66	8/15
Muffled	1.00	0.50	0.67	0/2
Repeated onset	0.63	0.39	0.48	1/7
Skipped words	0.77	0.55	0.64	–
Restarted phrase	0.89	0.52	0.65	–
All	**0.82**	**0.56**	**0.66**	**31/58**

P/R/F1: synthetic, leave-one-speech-out, 644 labels; 95 % bootstrap CI of F1 0.62–0.70. The original nine prosodic types on

their own clips: F1 0.70, P 0.88. Human R: performed flaws found, leave-one-speaker-out (reading errors were not scripted). F1 is 0.59 at tIoU ≥ 0.5 and 0.41 at 0.7. Matched flaws: mean tIoU 0.76, median boundary error 150 ms. Estimated severity is exact 61 % of the time.

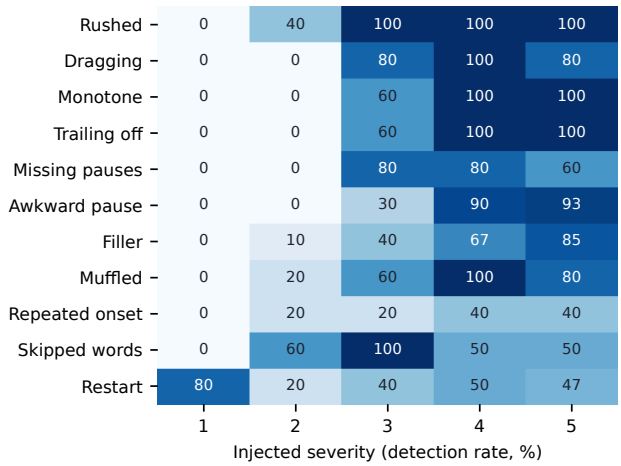


Figure 2. Detection rate rises with severity. Severity 1 sits inside natural take-to-take variation, so the gates deliberately leave most of it unreported.

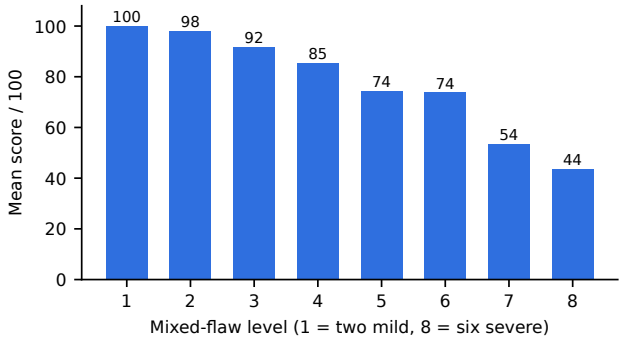


Figure 3. Mean score across the mixed-flaw gradient (Spearman ρ = −0.95; ρ = −0.79 between total injected severity and score over all synthetic clips). Unflawed controls score 98.4–100.

Human recordings and the listening audit

On team recordings, 53 % of performed flaws are found (95 % CI 39–68 %; by recorder F1 0.48, 0.23, 0.12, tracking how strongly each person acted the flaws). Measured precision is only 0.21, because any finding outside a scripted flaw counts as false, and people reading flaws on cue also slip unscripted. A blind listening audit tested this: a listener rated 30 snippets against the same words in the speaker's clean take, without knowing why each was picked. Unscripted findings were heard as worse delivery in 12 of 16 cases (95 % CI 50–90 %), scripted flaws in 6 of 7, and random spans with no finding in 0 of 7. Counting those as real puts precision near 0.80 (95 % range 0.61–0.92). The listener named the same problem as the detector in only 17 % of those cases: SpeechLens locates unscripted problems more reliably than it labels them. One listener from the team is an estimate, not a measurement.

Why the reference matters: ablations

Each component is switched off in turn on the same cached alignments and features (in-sample thresholds). The *reference-free baseline* keeps the whole detector but compares every passage with the speaker's own running median (±7 words) instead of a reference reading, the usual approach when no reference exists.

Variant	P	R	F1	Ctrl/min	Human P	Human R
Full system	**0.85**	**0.56**	**0.68**	**0.03**	**0.23**	**0.55**
No gates	0.72	0.66	0.69	0.03	0.10	0.62
No conflict rules	0.71	0.57	0.64	0.03	0.20	0.59
No habit/local split	0.85	0.56	0.67	0.03	0.20	0.47
No speaker scale	0.85	0.50	0.63	0.03	0.17	0.47
No reference	0.04	0.26	0.07	9.44	0.10	0.36
No ref., oracle-tuned	0.10	0.21	0.14	2.30	0.21	0.21

Without a reference, ordinary prosodic variation (a rhetorical pause, an emphatic slow phrase) looks like a flaw: even with thresholds tuned on the evaluation labels themselves, F1 is 0.14 and unflawed speech draws 2.3 findings per minute. The gates trade a little synthetic recall for twice the human precision; the habit/local split and the speaker-relative scale matter most on human readers, whose overall pace and range differ from the reference.

Counterfactual intervention

The explanation of each finding names one cause (articulation rate, pitch range, level, high-band energy, a pause, a held vowel, repeated fragments). We intervene on exactly that variable in the participant's own recording, moving it to the reference value while keeping the overall habit: PSOLA time-scaling or pitch-range widening, a per-word gain curve, a high shelf, or cutting / reinserting audio (for skipped words, the reference's own audio of them). The repaired recording then goes through the full pipeline again. Over 87 severity-3 and -4 single-flaw clips of all 11 types where the flaw was found, the finding disappeared in 91 % and was reduced or removed in 92 %; a different finding appeared nearby in 4.6 %. Mean score rose from 93.1 to 98.0 (median: 100 % of the gap to the clean reading recovered). A restart is repaired by cutting the second saying (7/7 removed), skipped words by splicing in the missing words from the reference (8/9). Muffled speech is the exception (1 of 8 removed): a low-pass filter destroys the high band, so boosting it cannot restore it. The same repair is offered in the dashboard as *Hear it fixed*.

Stress test

Only the participant's recording is degraded; the reference stays clean, as when a phone recording in a noisy room is judged against a good reading. Each condition uses 65 clips (all severity-4 single flaws of the 11 types, the level-6 mixed clips, and each clean speech as a null).

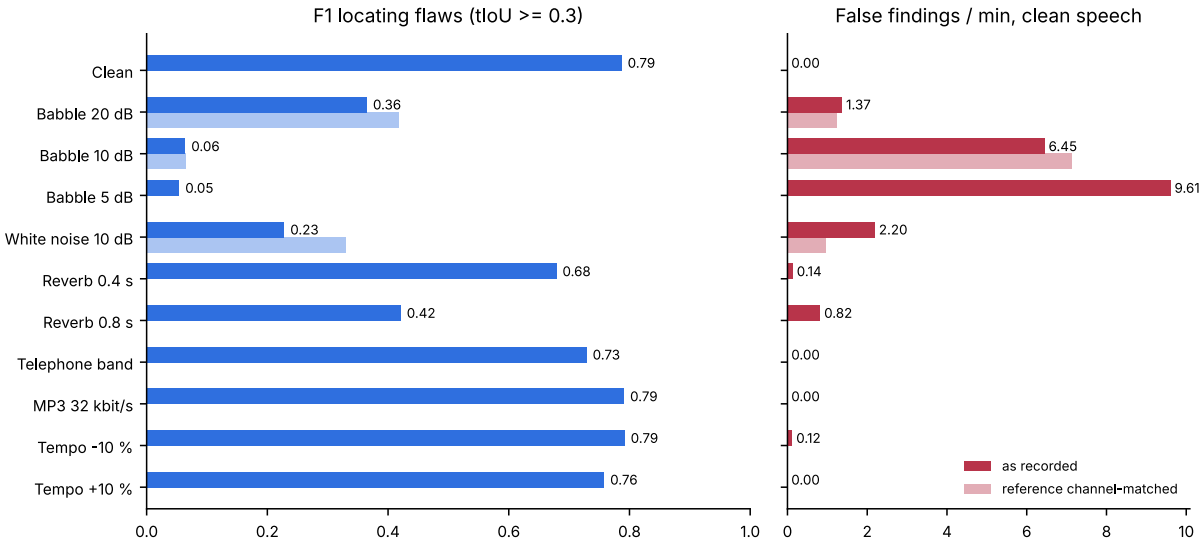


Figure 4. Grounding under degraded participant audio. Light bars: the reference put through the participant's estimated noise channel.

Against the clean condition (F1 0.79 on this subset), MP3 at 32 kbit/s (0.79), a uniformly 10 % faster or slower reading (0.79 / 0.76; the habit/local split absorbs it, word onsets move 7–9 ms), a telephone band (0.73) and moderate reverberation (0.68) barely matter. Background noise does: at babble SNR 20 dB F1 falls to 0.36 and clean speech draws 1.4 false findings per minute, and at 10 dB the comparison breaks down. Noise fills pauses and the quiet ends of words, which the contrast with a clean reference reads as shorter pauses, longer words and held sounds. Adding noise of the participant's estimated spectrum to the reference (*channel matching*) recovers part of it (babble 20 dB: 0.36 → 0.42; white noise 10 dB: 0.23 → 0.33). The app measures both noise floors relative to each speaker's level; above 6 dB of excess noise (none of our recordings) it applies channel matching and warns that timing and clarity findings may reflect the room.

New voices

All gold speakers are male presidents, so we built 20 new gold passages (about 60 s each) from LibriSpeech dev-clean readers the system never saw, **10 women and 10 men**, injected every flaw type at severity 4, and evaluated with every threshold, gate and severity table frozen. F1 is **0.85** (P 0.90, R 0.81; women 0.86, men 0.85) over 334 flaws; prosodic types transfer almost perfectly (rushed 0.98, dragging 1.00, monotone 0.97), repeated onsets (0.57) and skipped words (0.61) remain the weakest. Clean audiobook audio is easier than historic speech recordings, but the result shows the speaker normalisation holds across voices.

6. Dashboard and reproducibility

A FastAPI backend serves a no-build web dashboard. It accepts audio (WAV, MP3, M4A, OGG) and transcripts, or a dataset clip, and shows the score, six dimension bars, a zoomable time-aligned overlay of pitch, loudness and articulation (reference warped onto the participant's timeline through paired word boundaries), ground truth when known, clickable flaw bands, and per-finding detail with the measurement table, the rule that fired, A/B playback of the participant and the reference at the same words, and *Hear it fixed*, the counterfactual repair of that finding in the participant's own voice. Participants can upload a file or record from the microphone; a reference uploaded without a transcript is transcribed by greedy CTC decoding with the same acoustic model. Each analysis ends with a coach's summary (priorities with drills, a voice-quality table, a printable report); *Practice this part* records a retake of one passage and compares it with the original passage against the same reference crop, and retakes of a text are tracked as a progress chart. A hosted version of the dashboard with every dataset clip pre-analysed runs on GitHub Pages.

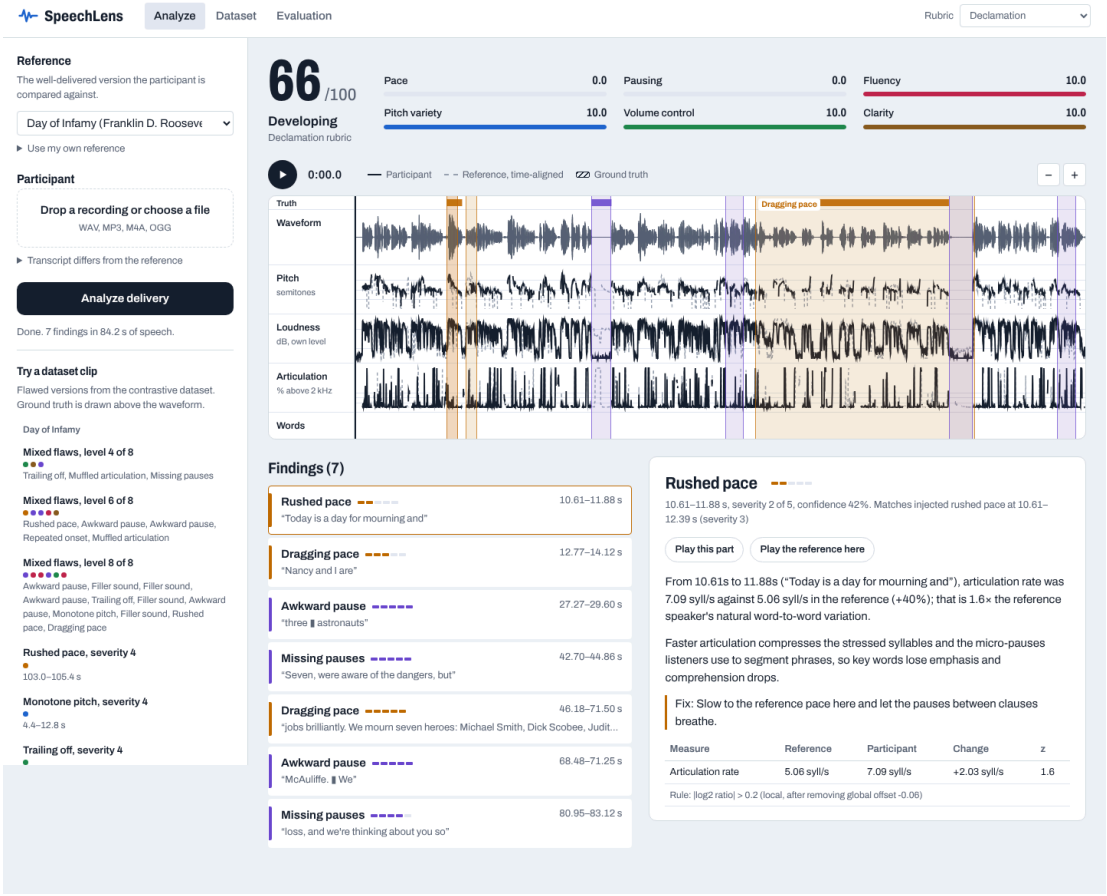


Figure 5. A team recording against the speaker's own clean take: the scripted fast sentence, the pause after "three" and the slowed list of names are located and explained.

Reproducibility. CPU only; pinned dependencies; Docker image; `reproduce.sh` rebuilds the dataset and every number from the raw speeches; analyses are cached by content hash; a re-run in a fresh process gives byte-identical output (checked inside the evaluation). 25 unit tests cover the Viterbi, text normalization, injection labels, determinism, audio splicing, channel matching and the recording check.

7. Limitations

- The reference must read the same text; the aligner is English-only; accents and spontaneous speech are untested.
- Background noise the reference lacks breaks the comparison (stress test); channel matching only partly compensates, so the app warns.
- Synthetic flaws are cleaner than human ones; human results are reported separately and with 58 verified flaws, so their intervals are wide.
- Whispering is not modelled as a flaw, and phone gain control hides fades.
- Muffled articulation is measured through high-frequency energy, so a channel change mid-recording can mimic it.
- Labels for unscripted findings are less reliable than their locations.

Code, dataset and documentation: GitHub repository (README). Speech audio and transcripts: Miller Center, University of Virginia (public domain). Alignment model: facebook/wav2vec2-base-960h (Apache-2.0), ONNX export by Xenova. Praat via Parselmouth (GPL-3.0).